

Mia Francis Bonini

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SUMMARY

PhD engineer with broad expertise in computational modeling, medical device evaluation, and clinical research. Experienced building and validating physics-based simulations against real-world experimental and clinical data and leading research studies from IRB submission through publication. Skilled in Python, MATLAB, and R for data analysis and automation, with an extensive record of peer-reviewed publications, technical presentations, and cross-functional collaboration.

SKILLS

Imaging & Ultrasound: ultrasound/echocardiography, CT, MRI, image segmentation and 3D anatomical reconstruction

Clinical Collaboration & Training: direct partnership with cardiac surgeons, interventional radiologists, and clinical staff; case-based technical support; training and mentoring; technical presentations to clinical and multidisciplinary audiences

Cross-Functional Coordination: collaboration across engineers, clinicians, and research teams; feedback loops with R&D on device performance; management of multiple concurrent projects and deadlines

Computer: Microsoft Word, Excel, PowerPoint, Outlook; Python, MATLAB, linear regression, random forest, neural networks

EXPERIENCE

Research Staff | *University of Michigan — Cardio Biomechanics Lab* 11/2025 – present

- Leading development of a physics-based computational modeling framework to simulate complex biological systems and evaluate medical device performance, with a focus on cardiovascular applications.
- Translating clinical and imaging data into quantitative models that inform device design, treatment strategy, and risk assessment ahead of clinical trials.

Graduate Student Research Assistant | *University of Michigan — Cardio Biomechanics Lab* 08/2020 – 11/2025

- Expertise in data driven CFD and FSI modeling.
- Developed patient-specific anatomical models from clinical imaging data (CT, MRI, ultrasound) and validated simulation outputs against clinical measurements, achieving <10% error
- Defined user needs and device performance requirements through direct collaboration with physicians and engineers, translating clinical questions into actionable research and design criteria
- Designed and executed in-silico feasibility studies, concept studies, and technology assessments to support novel medical device development, including a \$415K funded program and an associated patent application
- Built and maintained clinical data pipelines integrating large-scale imaging and electronic health record (EHR) data, applying statistical and machine learning models (Python, Matlab) for outcome prediction
- Contributed to 4 IRB submissions and completed human-subjects research compliance training, ensuring data integrity and regulatory compliance throughout
- Led multiple concurrent research projects to on-time completion; authored 12+ peer-reviewed publications and delivered 7+ international presentations
- Mentored 14 junior researchers in computational methods, data analysis, and scientific communication

Undergraduate Researcher | *Wake Forest University — Dept. of Biomedical Engineering* 06/2019 – 08/2019

- Built an automated pipeline to segment MRI images and analyze muscle fat concentration for a NASA-funded spaceflight study, while maintaining HIPAA-compliant datasets.
- Extracted spinal bone and muscle tissue and performed mechanical compression testing to characterize tissue properties.

Engineering Intern — SSD & Thermal Solutions | *Intel Corporation* 06/2018 – 08/2018

- Characterized thermal and mechanical performance of device assemblies using wind tunnel and still-air chamber testing, analyzing temperature profiles.
- Standardized laboratory testing protocols and documentation to reduce operator variability and improve reproducibility.

EDUCATION

Ph.D. in Biomedical Engineering — University of Michigan 11/2025

M.S. in Biomedical Engineering — University of Michigan 05/2022

B.S. in Mechanical Engineering — University of Portland 05/2020

HONORS & AWARDS

1. Functional Imaging and Modeling of the Heart Early-Career Scholarship Award 06/2025
2. Marvin Kirsh Resident Research Symposium, 1st Place Basic Science 03/2025
3. U.S. Association for Computational Mechanics Travel Award 07/2024
4. University of Michigan Rackham Graduate Student Research Grant 11/2023
5. EUROMECH Young Researcher Award 07/2022